IO SKY

AI SCAN ENGINE & UNIVERSAL ASSESSMENT SYSTEM

Master Functional, Intelligence, Conversation, Evidence, Review, Report, Executive Briefing, Security, Operations & Publication Specification

Field	Value
Document status	Freeze Candidate v1.0
Authority	IO SKY Product Owner
Applies to	Operations Scan · Cyber Scan · Elite Scan
Classification	Proprietary / Confidential
Purpose	Authoritative universal implementation, validation and acceptance specification
Visual status	Functional/behavioural freeze candidate; report typography/tokens freeze only after mandatory reference-composition approval

Core principle: IO SKY specifies what the product must do, how it must behave, what users must experience, which controls must exist, and how compliance is accepted. The development team determines the engineering implementation, provided the resulting system demonstrably satisfies this specification.

0. Document Control, Authority & Interpretation

GEN-001	This document is the authoritative universal product specification for the shared IO SKY AI Scan Engine. It governs purchase, account binding, Shared Intake, assessment intelligence, memory, evidence, findings, review, reporting, publication, security, failure handling, operations and acceptance.
GEN-002	MUST/SHALL = mandatory; MUST NOT/SHALL NOT = prohibited; SHOULD = expected unless a documented exception is accepted by IO SKY; MAY = permitted.
GEN-003	The Universal Master defines HOW every AI Scan behaves. Separate Operations, Cyber and Elite methodology specifications define WHAT each product investigates.
GEN-004	Product methodologies MUST NOT override universal rules for adaptiveness, memory, evidence, confidence, review, publication, security, translation, report integrity or versioning.
GEN-005	Developers retain engineering freedom over frameworks, service boundaries, database implementation, queueing, internal enums, libraries and provider integration patterns, but only where resulting behaviour remains compliant.
GEN-006	No mandatory behaviour may be silently removed, weakened, merged or replaced because another implementation is easier, cheaper, already exists, or is preferred by the development team.
GEN-007	If a genuine contradiction, technical impossibility, security conflict or material ambiguity is identified, development of the conflicting behaviour MUST stop and the matter MUST be raised with IO SKY before an alternative is implemented. Silence is not approval.

Non-Interpretation Rule. Absence of a prescribed framework, library, table name, provider or internal enum does not permit simplification of required product behaviour. Existing implementation behaviour does not override this specification.

1. Product Definition & Quality Standard

PRD-001	The AI Scan is a premium adaptive professional assessment and decision-support product. It MUST NOT function or feel like a static questionnaire, survey, lead-generation quiz, fixed question tree, maturity calculator, generic chatbot, long form or questionnaire followed by an automatically generated PDF.
PRD-002	The customer value chain is: Understand → Investigate → Challenge → Validate → Deepen → Connect → Assess → Prioritise → Recommend → Human Review → Publish.
PRD-003	The system must be capable of establishing what is happening, what matters, why it matters, what may cause it, what is symptomatic, what works, what should remain unchanged, what deserves attention, what remains uncertain, what depends on something else, what deserves priority and what investment direction is or is not justified.
PRD-004	Valid outcomes include Preserve, Improve, Configure/Optimise, Integrate, Automate, Targeted Change, Investigate/Validate, Larger Transformation and No Material Intervention.
PRD-005	The Engine MUST NOT be designed to conclude by default that the customer should buy IO SKY, implement AI, automate everything, replace software, build custom software or undertake a transformation.
PRD-006	The system MUST NOT invent overall scores, maturity scores, readiness percentages, arbitrary efficiency percentages, company-health scores, fake confidence percentages or arbitrary weighted totals.
PRD-007	Confidence states are qualitative: High, Moderate, Low, Insufficient.

PRD-008 Customer-facing priority states are: Address First, Address Next, Planned Improvement, Monitor/Preserve, Further Investigation.

2. Commercial Product Configuration

Product	Price	Included	Optional	Delivery target
Operations Scan	€795 excl. VAT	Operations Assessment Report; Priority Roadmap; Executive Summary	Live Expert Review €295; Executive Briefing €395; Both €595	5–7 business days after submission
Cyber Scan	€1,595 excl. VAT	Cyber Assessment Report; Security Roadmap; Executive Summary	Live Expert Review €295; Executive Briefing €395; Both €595	7–10 business days after submission
Elite Scan	€3,295 excl. VAT	Elite Assessment Report; Strategic Priority Roadmap; Executive Summary; Live Expert Review Session	Executive Briefing €395	10–14 business days after submission

COM-001 There is no free AI Scan product.

COM-002 Add-ons are selected before payment and are not preselected. Operations/Cyber may select none, one or both. Elite may select only Executive Briefing because Live Expert Review is included.

COM-003 Elite is broader and cross-domain but MUST NOT be positioned or implemented in a way that implies Operations or Cyber are superficial or incomplete.

3. Universal Customer Lifecycle

Canonical lifecycle: Product Selection → Purchase Configuration → Organisation & Billing → Shared Intake → Audience → Publication Language → Full Review → Payment → Server-Side Payment Verification → New Account Creation OR Existing Account Binding → Entitlement → Assessment Creation → Adaptive Assessment → Submission → Analysis → Findings → Relationships → Priorities → Recommendations → Roadmap → Canonical Report → NL + EN Review Renderings → Automated Review Intelligence → Reviewer Assignment → Expert Review → Expert Approval → Super Admin Publication Review → Customer-Language Rendering → Executive Briefing where applicable → Final Validation → Publish → Correct Client Portal → Notification → Immutable Published Version.

LIFE-001 The lifecycle above is the universal business sequence. Technical implementation may be asynchronous, but it MUST preserve the defined state dependencies and publication gates.

LIFE-002 Customer-facing status labels may simplify internal states but MUST NOT misrepresent whether analysis, review or publication is complete.

4. Public Account Creation Boundary

ACC-001 IO SKY MUST NOT provide unrestricted public Client account registration.

ACC-002 A new Client account may originate only through an authorised IO SKY invitation/internal provisioning process or a successfully paid public AI Scan purchase.

ACC-003 The AI Scan purchase route is the sole public non-invitation self-service route for creating a new IO SKY Client account.

ACC-004 A generic /signup, login-registration path or equivalent endpoint MUST NOT permit unrestricted account creation. This boundary MUST be enforced server-side, not merely hidden in the interface.

ACC-005 A browser success redirect, entered email address, organisation name or Assessment ID is not sufficient proof to create, activate or bind a Client account.

5. New Customer Purchase & Account Journey

Required journey: Select Scan → Configure Add-ons → Organisation & Billing → Shared Intake → Report Audience → Report Language → Full Review → Stripe Payment → Server verifies payment → Account Activation Eligibility → Secure Account Creation/Activation → Organisation/Purchase/Entitlement/Assessment binding → Client Portal → Continue Assessment.

ACC-006	No active Client account is created before confirmed payment through the public AI Scan route.
ACC-007	Successful payment verification creates eligibility for secure account activation; it does not create an anonymous usable account.
ACC-008	The backend MUST bind User Account → Organisation Membership → Verified Purchase → Entitlement → Assessment ID → Pre-Purchase Assessment Snapshot.
ACC-009	The customer MUST NOT manually claim a purchase by entering an Assessment ID.
ACC-010	Shared Intake and other successfully captured pre-purchase assessment context MUST NOT be requested again after account activation.
ACC-011	A successfully paid purchase MUST survive browser closure or account-activation failure in a recoverable Payment Confirmed — Account Activation Pending state.
ACC-012	Another person MUST NOT be able to claim that purchase merely by knowing email, organisation name, Assessment ID or a browser URL.
ACC-013	Recovery from Account Activation Pending MUST require secure proof appropriate to the activation process.

6. Existing Client Journey, Prefill & Snapshot Control

ACC-014	An authenticated existing Client who starts or purchases a Scan remains authenticated and MUST NOT be unnecessarily routed through new-account registration.
ACC-015	The system MUST NOT create a duplicate Client account or duplicate Organisation merely because the existing Client purchases a new Scan.
ACC-016	Eligible known Organisation/account information SHOULD be automatically prefilled, including Organisation Name, country, registration identifier, VAT/Tax ID and other appropriate known information.
ACC-017	Prefilled values MUST remain visible to the customer and editable for the current purchase/assessment where that field is permitted to be edited.
ACC-018	Account/Organisation Master Data and the Assessment Snapshot are separate authoritative concepts.
ACC-019	Editing a prefilled value for the current Scan MUST NOT silently overwrite the authoritative Client or Organisation master record.
ACC-020	A separate explicit profile-update operation is required to change master data.
ACC-021	Previous assessment answers MUST NOT be copied into a new Scan as though the customer answered them again.
ACC-022	Historical assessment answers remain associated with their historical Assessment ID and Methodology Version.
ACC-023	An unauthenticated person entering an email matching an existing account MUST NOT be shown protected account or organisation information.

ACC-024 A new purchase MUST NOT be bound to an existing Client solely because the same email string was entered. Existing-account ownership must be authenticated before protected data is disclosed or binding occurs.

ACC-025 Account-collision handling MUST prevent duplicate accounts, accidental takeover, wrong Organisation membership and cross-client purchase binding.

7. Organisation, Billing & Shared Intake

INTK-001 Collect Organisation Name, Country of Registration, Company Registration Number, and VAT/Tax ID where applicable. Company Registration Number and VAT/Tax ID are separate concepts.

INTK-002 Organisation Name is collected once and reused throughout purchase and assessment; it MUST NOT be needlessly asked twice.

INTK-003 Country-aware validation/formatting MUST support the international nature of IO SKY. The implementation MUST NOT assume every organisation has a Dutch KVK number.

INTK-004 The production validation architecture MUST document supported country/identifier behaviour, provider/source, verification status, provider unavailability, invalid format, failed verification, fallback and stored validation result. IO SKY approves the production approach.

Shared Intake introduction:
Before we begin
Tell us a little about your organisation and what you want to understand or decide. This gives us the context to shape the assessment around your situation, without assuming the outcome in advance.

Field	Customer prompt/options	Requirement	Validation
Organisation context	Briefly describe what your organisation does, who it serves, and its main products or services.	Required	Free text
Organisation size	Just me; 2–10; 11–50; 51–200; 201–500; 501–1,000; 1,001+	Required	Single select
What prompted assessment?	Clearer current situation; improvement greatest impact; specific concern; overlooked risks/opportunities; make/validate important decision; Other	Required	Max 2; Other requires text
What should assessment help understand/decide?	Customer's objective	Required	Free text
Considering significant change/investment?	Yes / No / Not sure	Required	If Yes, description required
Report Audience	Five controlled profiles	Required	Max 2; if 2, choose Primary
Report Language	Supported IO SKY publication languages	Required	Single select

INTK-005 Q1–5, Audience and Report Language are required. Other requires supporting text. Significant Change/Investment requires top-level selection and description when Yes.

INTK-006 Shared Intake is assessment context, not a public explanation of IO SKY methodology. Do not add per-question 'Why we ask' copy unless separately approved.

8. Report Audience & Language

Audience	Typical readers	Communication emphasis
Executive Leadership	CEO, Founder, Managing Director, General Management	Business consequences, decisions, priorities, investment, dependencies, outcomes
Operations Leadership	COO / Operations	Process, execution, capacity, bottlenecks, ownership, sequencing
Technology & Security Leadership	CTO, CIO, CISO, IT	Architecture, systems, data, integrations, controls, dependencies, risk
People & Organisational Leadership	HR / People / organisational leadership	Roles, ownership, adoption, ways of working, capacity, organisational effects
Board & Investors	Board, shareholders, investors, advisers	Strategic meaning, material risks, resilience, scalability, dependencies, governance

AUD-001 Maximum two audiences. If two are selected, Primary Audience is mandatory and Secondary Audience is the other.

AUD-002 Audience changes communication emphasis, terminology and explanatory depth only. It MUST NOT change findings, confidence, priority, recommendation, roadmap, facts or conclusions.

AUD-003 Do not build separate independent reports for audience combinations. Primary drives dominant communication layer; Secondary adds relevant context.

LNG-001 Customer Publication Language is selected from the centrally supported IO SKY language configuration. Do not maintain a conflicting second hard-coded list.

LNG-002 Customer Publication Language and Expert Review Language are separate concepts.

9. Pre-Payment Review & Payment

PUR-001 Before Continue to payment, show a structured review of the full information entered: organisation, registration/tax data, Scan, add-ons, Shared Intake, contemplated investment, audiences, Primary Audience, language, subtotal, tax and total.

PUR-002 The customer MUST be able to return and edit eligible information before payment.

PAY-001 Payment status is server-authoritative. A browser success redirect is never authoritative proof of payment.

PAY-002 Payment/webhook processing MUST be idempotent. Duplicate events MUST NOT create duplicate orders, entitlements, assessments, accounts or one-time business notifications.

PAY-003 The IO SKY record must retain sufficient Stripe/payment references for reconciliation while avoiding unnecessary storage of sensitive payment data.

PAY-004 The linkage must be reconstructable: Client → Organisation → AI Scan → Assessment ID → Product/Add-ons → Billing Record → payment references → amount/tax/currency/status → Entitlement → timestamps.

10. Assessment ID, Methodology & Configuration Versioning

DATA-001 Every Scan receives a permanent unique Assessment ID linking Client, Organisation, Purchase, Entitlement, Product, Methodology Version, Assessment, Review, Report, Published Version, Executive Briefing and relevant Audit History.

DATA-002 Authorised internal users must be able to locate an assessment by Assessment ID and by Organisation/company name where their permissions permit.

DATA-003	Every Assessment stores the Methodology Version and relevant configuration/version references that governed it.
DATA-004	Historical assessments MUST NOT retroactively change when current methodology, wording, product configuration or report configuration changes.

11. Assessment UX & Interaction Model

UX-001	The Assessment Experience belongs visually to the same IO SKY design family as the website and Client Portal. It MUST NOT look like a generic survey tool or chatbot.
UX-002	Dominant assessment background is Deep Teal #0D2D2E. Primary actions and active/progressive states use IO SKY Orange #F58A1F in a controlled manner. Primary assessment text is off-white. Grey typography/UI MUST NOT be used as the default hierarchy mechanism.
UX-003	The experience is minimal, spacious, calm, controlled and premium. One meaningful interaction or context group receives sufficient visual space; do not present a giant question list.
UX-004	Exact font family, typography scale, weights, line heights and responsive typography are NOT frozen by this Master. They must be visually evaluated against the final website/portal typography, approved by IO SKY, then derived into implementation tokens.
UX-005	A meaningful visual progress indicator MUST exist. It becomes visually stronger in IO SKY Orange as genuine assessment progression occurs.
UX-006	Because the assessment is adaptive, progress MUST NOT claim a fabricated exact percentage or fixed 'questions remaining' count unless a future implementation can prove that value is truthful.
UX-007	Where an interaction has one clear valid completion state, the interface SHOULD automatically advance after the input has been successfully validated and safely persisted. The customer should not need to press Next after every simple interaction solely to advance.
UX-008	Automatic progression MUST NOT occur before required validation and persistence are complete, MUST NOT skip required information, and MUST NOT create uncertainty about whether an answer was saved.
UX-009	The customer MUST retain navigation control. A visible Back action must permit return to an earlier editable interaction where assessment state permits. A Next/Continue action remains available where explicit confirmation is useful or auto-advance is inappropriate.
UX-010	If the customer returns and materially changes an earlier answer, the system MUST update/invalidate affected downstream assessment state as appropriate. It MUST NOT continue relying on superseded information as if unchanged.
UX-011	Automatic progression is a UX acceleration mechanism, never a bypass of validation, persistence, adaptiveness or state integrity.

12. Adaptive Intelligence & Conversational Behaviour

Core intelligence cycle: Understand → Investigate → Challenge → Validate → Deepen → Connect → Assess → Prioritise → Recommend.

AI-001	After every relevant answer the Engine must: understand communicated meaning; contextualise it against persistent state; check consistency; assess relevance; assess uncertainty; assess materiality; decide depth; choose follow-up/continue/flag; update structured assessment state.
AI-002	The Engine MUST NOT ask an extra question merely because one is available. Every follow-up must have an assessment purpose.
AI-003	Legitimate follow-up purposes include clarification, contradiction resolution, impact/materiality, cause, dependency, risk, opportunity, confidence improvement, strength preservation and decision relevance.
AI-004	The Engine must use a hybrid architecture: controlled base objectives/questions/triggers/follow-up/depth/stop metadata plus bounded dynamic formulation where needed for ambiguity, contradiction or a material investigation path.
AI-005	The customer experience must feel like an intelligent professional conversation, not a rigid questionnaire and not a gimmicky open-ended chatbot.
AI-006	The Engine must be capable of challenging a customer's proposed solution without assuming the customer is wrong. It investigates the underlying situation before treating the proposed solution as necessary.
AI-007	The Engine must recognise positive evidence and strengths. A strength can trigger further investigation where understanding why it works materially improves the assessment.
AI-008	The Engine must accept 'I don't know' or equivalent uncertainty as valid information. It MUST NOT force guessing.

13. Persistent Assessment Memory

MEM-001	Relevant information from the beginning of the assessment remains available throughout the assessment, including after save/resume.
MEM-002	The system MUST NOT rely only on a short rolling model/chat context window for assessment continuity.
MEM-003	Structured state must be maintained for Organisation Context, Objectives, Decision Context, Contemplated Investment, Reported Facts, Customer Assumptions, Terminology, Signals, Strengths, Constraints, Risks, Opportunities, Open/Resolved Investigation Paths, Contradictions, Unknowns, Candidate Findings, Relationships, Confidence and Reviewer Attention.
MEM-004	Customer-visible conversation and Internal Assessment State are separate.
MEM-005	Raw private model chain-of-thought is neither required nor to be exposed. Developers must instead maintain observable structured state/decision semantics sufficient to validate behaviour.
MEM-006	Permitted observable state semantics include SIGNAL_CREATED, SIGNAL_STRENGTHENED, CONTRADICTION_DETECTED, CLARIFICATION_REQUIRED, INVESTIGATION_DEEPENED, PATH_SUFFICIENT, UNKNOWN_RECORDED, STRENGTH_ESTABLISHED, CANDIDATE_FINDING_CREATED, CONFIDENCE_UPDATED and REVIEWER_ATTENTION_REQUIRED.

14. Depth Engine

DEP-001	Continue investigation while more information can materially improve reliability, relevance or actionability. Stop when further information is unlikely to materially change the assessment.

DEP-002 Logical depth states include Surface, Clarification, Investigation, Deep Investigation, Sufficiently Investigated, Unresolved and Reviewer Attention.

DEP-003 Potential deepening triggers include high impact, unclear information, low confidence, contradiction, new material path, root-cause uncertainty, decision relevance, material risk/opportunity, dependency and material strength.

DEP-004 Stop conditions include sufficient understanding, stable conclusion, low materiality, outside scope, unavailable information, specialist investigation required, or uncertainty that should responsibly remain.

DEP-005 There is no arbitrary maximum-three-follow-ups rule. Equally, the Engine MUST NOT continue questioning without material purpose.

DEP-006 Branch priority is driven by impact, uncertainty, decision relevance, risk/opportunity and relationships/dependencies rather than by a fixed question count.

15. Evidence, Confidence & Contradictions

Evidence state	Meaning	Required treatment
Reported	Customer states it	Not independently verified by that fact alone
Corroborated	Multiple relevant assessment elements support it	Stronger basis, still bounded by assessment context
Derived Observation	Directly derived from available assessment information	Must remain traceable
Inference / Hypothesis	Plausible but not sufficiently established	Use qualified language
Conflicting	Relevant information conflicts	Clarify or preserve conflict
Unknown / Insufficient	Not enough information	Do not manufacture certainty

EVD-001 Customer-reported information MUST NOT silently be represented as independently verified fact.

EVD-002 Normal Scan operation MUST NOT create a routine customer evidence-upload/document-request phase. The primary basis is Shared Intake, answers, adaptive follow-ups, cross-checks, consistency, contradictions and cross-answer relationships.

CNF-001 Confidence is per finding and qualitative: High, Moderate, Low or Insufficient. No fake confidence percentages.

CNF-002 Confidence may change based on clarity, completeness, consistency, corroboration, specificity, contradiction, unknowns, depth and inference distance.

CNF-003 Recommendation certainty MUST NOT exceed the confidence justified by the underlying finding.

CTR-001 Contradiction categories include Direct, Partial, Contextual, Temporal, Definition and Unresolved.

CTR-002 Potential contradictions must be neutrally clarified when clarification can reasonably resolve them; contradiction handling must not become adversarial questioning.

CTR-003 Material unresolved contradiction may reduce confidence, prevent establishment, create Further Investigation and/or create Reviewer Attention.

16. Finding Lifecycle & Cross-Finding Intelligence

FND-001 An answer, unusual answer, signal or model hunch is not automatically a Finding.

FND-002	Finding development follows: Assessment Information → Signals → Related Signals → Possible Issue/Strength/Opportunity → Contributing Factors → Root-Cause Hypothesis → Business Consequence → Candidate Finding → Established/Limited/Hypothesis/Unable to Establish.
FND-003	The system distinguishes symptom, contributing factor, root cause/likely root cause, consequence, strength, opportunity, constraint and risk/exposure.
FND-004	Finding end states are Sufficiently Established; Established with Limitation; Hypothesis — Further Investigation Required; Unable to Establish.
REL-001	The system must identify material relationships including shared cause, cause/consequence, dependency, reinforcement, multiple symptoms of one issue and mitigating strength.
REL-002	Related findings may form Finding Clusters. The system must avoid presenting multiple symptoms as unrelated problems when evidence supports a common underlying relationship.

17. Priority, Recommendation, Decision & Roadmap

PRI-001	Priority considers impact, urgency, confidence, dependencies, risk/opportunity, decision relevance and implementation logic.
PRI-002	Priority MUST NOT be reduced to a 1–10 or 0–100 score.
REC-001	No recommendation may prescribe a larger intervention than the findings justify.
REC-002	Each material recommendation must be traceable to supporting finding(s) and establish what, why, intended outcome, first step, dependencies, certainty and intervention type.
REC-003	Decision principles: configure/optimise before replacement where sufficient; integrate before rebuild where sufficient; improve a broken process before automating it; investigate before unsupported investment; preserve what works; custom/new intervention only when justified.
REC-004	Permitted investment-direction states include No material investment currently justified; Optimise/configure first; Integrate/automate targeted areas; Targeted replacement may be justified; Larger transformation may be justified; Validate further before investment.
RDM-001	Roadmap structure is NOW → NEXT → THEN → LATER/MONITOR. Parallel tracks are allowed where capacity and dependencies permit.
RDM-002	Roadmap items may include action, rationale, intended outcome, supporting findings, dependencies, priority, sequencing rationale, indicative complexity and relevant ownership/expertise.

18. Autosave, Resume, Back Navigation & Submission

SAV-001	Autosave must persist answers plus sufficient structured assessment state, context, investigation paths, signals, contradictions and progression to resume coherently.
SAV-002	UI distinguishes Saving, Saved and Save failed. It MUST NOT falsely display Saved.
SAV-003	Resume restores the last confirmed coherent assessment state, not merely a list of answers.

SAV-004	When an earlier answer is changed through Back navigation, dependent state must be recalculated, invalidated or re-investigated as required. Superseded data must not remain silently authoritative.
SUB-001	Final customer action is Submit Assessment.
SUB-002	Submission creates an immutable Submitted Snapshot.
SUB-003	Duplicate submit attempts are idempotent and create one logical submission.

19. Internal Post-Submission Flow

Authoritative internal chain: Submitted Snapshot → Analysis → Findings Registry → Relationship/Root-Cause Map → Evaluation/Confidence → Priority → Recommendations → Decision Considerations → Roadmap → Structured Canonical Report → NL + EN Review Renderings → Automated Review Intelligence/QA → Reviewer Assignment → Expert Review.

ANA-001	No incomplete AI output is automatically visible to the customer.
ANA-002	AI/provider failure preserves the Submitted Snapshot and recoverable processing state; it MUST NOT create conflicting duplicate canonical versions.

20. Expert Review Language, Assignment & Ownership

REV-001	Before Expert Review, controlled Dutch and English internal review renderings must exist from the same canonical assessment. They are not independent analyses.
REV-002	Reviewer chooses Dutch or English as review language. Material reviewer edits modify canonical content; dependent renderings are then refreshed.
REV-003	Every Scan has an Assigned Reviewer. Launch assignment is controlled/manual by Super Admin.
REV-004	Reviewer requires AI Scan Review permission, correct product/scope and explicit assignment. Role alone does not grant access.
REV-005	Eligible assigned reviewers may be Developer, Admin, Super Admin or another future authorised internal role with the required permission/scope.
REV-006	Review access does not transfer ownership. The assessment remains owned/bound to the purchasing Client/Organisation and appears only in the reviewer's internal review workspace.
REV-007	Assignment and reassignment are audited and generate appropriate internal tasks/notifications.

21. Expert Review Workspace & Review Brief

REV-008	Workspace header shows Product, Client/Organisation, Assessment ID, Methodology Version, completion date, Primary/Secondary Audience, Customer Publication Language, Review Language, status and review progress.
REV-009	Workspace provides two linked modes: Report View and Reviewer Intelligence Panel.
REV-010	Expert Review Brief identifies, where relevant: Executive Review Brief, Areas Requiring Human Attention, High-Attention Findings, material Moderate/Low-confidence findings, contradictions, root-cause hypotheses, cross-finding relationships, high-impact recommendations, investment-sensitive recommendations, potential overstatement, unknowns/limitations, language attention and automated consistency flags.

REV-011 Review Brief directs human attention; it does not replace full Expert Review.

- Factual fidelity
- Reasoning integrity
- Finding validity
- Symptom vs root cause
- Contradictions/unknowns
- Confidence calibration
- Business impact/materiality
- Cross-finding relationships
- Recommendation proportionality
- Investment decision implications
- Priority logic
- Roadmap sequencing/dependencies
- Strengths/what to preserve
- Audience/language quality insofar reviewer is competent in review language
- Overall completeness/internal consistency

22. Finding Inspector, Reviewer Actions & Change Control

REV-012 Finding Inspector exposes, where applicable: Finding, Assessment Basis, relevant answers/follow-ups, Evidence State, Related Findings, Root-Cause Reasoning, Confidence/rationale, Contradictions/Unknowns, Business Impact, Priority/rationale, Proposed Recommendation and Roadmap Position.

REV-013 Customer source answers are immutable after submission. Reviewer changes affect derived objects, never historical source answers.

REV-014 Reviewer actions include controlled Edit, Request Targeted Regeneration, Return for Re-analysis, Mark Reviewed, Add Internal Review Note, Complete Review and Approve & Send to Super Admin.

REV-015 Editorial/non-material edits are logged. Material edits to finding, confidence, root cause, impact, priority, recommendation, roadmap, certainty or dependencies require a rationale.

REV-016 If the underlying reasoning is materially wrong, the reviewer must use Return for Re-analysis rather than silently rewriting the result as if the original analysis were valid.

REV-017 No destructive overwrite. Preserve Original AI Version → Reviewer Edited Version → Change Type → Reviewer → Timestamp → Reason where required.

REV-018 Targeted regeneration must preserve unaffected approved content and rerun downstream consistency checks for affected relationships, priorities, recommendations, roadmap, Executive Summary, language renderings and Executive Briefing.

REV-019 AI regeneration MUST NOT silently restore content previously rejected or materially corrected by the reviewer.

23. Review Completion, Handoff & Versioning

REV-020	Review completion checks unresolved critical flags, required review areas, blocking contradictions, low-confidence high-impact recommendations, missing roadmap relationships, semantic inconsistencies, incomplete sections and pending regeneration.
REV-021	Critical integrity/security blockers cannot be bypassed. Any future permitted non-critical exception requires Super Admin reason and audit.
REV-022	Reviewer handoff summary covers what was reviewed, what required attention, what was incorrect/insufficient, what changed, why, what regenerated/reanalysed, remaining limitations and points for Super Admin.
VER-001	Version sequence supports Draft v1 — AI-generated; Review Revision(s); Approved v1; Published v1.
VER-002	Any material content change after approval invalidates approval and requires reapproval.
VER-003	Published versions are immutable.

24. Super Admin Publication Package & Gate

PUB-001	Super Admin package shows Client/Organisation, Scan Type, Assessment ID, Methodology Version, Reviewer, review timestamps, Approved Version, Customer Publication Language, audiences, deliverables, Expert Review Summary, automated Change Summary, unresolved limitations and publication blockers.
PUB-002	Super Admin actions: View Final Report, View Review Brief, View Renderings, View All Changes, View Reviewer Notes, View Assessment Reasoning/Basis, Return to Reviewer, Publish Assessment.
PUB-003	Pre-publish gate requires Expert Review completed, critical flags zero, language/semantic/rendering QA passed, correct client/organisation binding, correct entitlement/deliverables, and Executive Briefing ready or not applicable.
PUB-004	If Super Admin returns the assessment, a mandatory note is required, reviewer task reopens and previous approval is invalid for modified content.
PUB-005	Four-eyes separation is preferred. Super Admin may self-review where authorised, but review completion and Publish remain explicit separate actions; self-review MUST NOT automatically publish.

25. Publication Binding, Atomicity & Delivered Date

PUB-006	Publication destination is fixed server-side from Assessment ID → Purchase/Order → Client Account → Organisation → Entitlement → Approved Publication Version → Customer Publication Language → Deliverables. Frontend never determines the destination.
PUB-007	Publication must validate all binding relationships server-side before exposure. Wrong-report/wrong-portal publication is a critical security/privacy failure.
PUB-008	Publication is atomic from the customer perspective: no report visible without correct entitlement, no PDF from wrong version, no notification claiming Ready before valid publication, and no Executive Briefing linked to the wrong source version.
PUB-009	If publication fails, record recoverable failure and do not expose a partial customer deliverable.

PUB-010	Successful publication timestamp determines the report Delivered date. Purchase, submission, generation, review and approval dates MUST NOT be used as the Delivered date.
PUB-011	Delivered date is system-derived from successful publish and is not independently manually editable.

26. Translation & Customer-Language Pipeline

Pipeline: Approved Canonical Content → Professional Language Rendering → Terminology Control → Semantic Equivalence Validation → Language Quality Validation → Layout Validation → Customer Publication Version.

LNG-003	Translation/rendering performs no new assessment, reasoning or conclusion generation.
LNG-004	Language may change sentence structure and terminology appropriately, but MUST NOT change meaning, certainty, findings, confidence, priority, recommendations, roadmap, dependencies, limitations or commercial/decision implication.
LNG-005	Raw machine translation is not acceptable as final published quality.
LNG-006	Controlled terminology and protected terms must be supported. Examples of semantic drift that are prohibited include changing 'likely' to certain, 'investigate' to implement, or changing a concept such as technology into infrastructure where meaning changes.
LNG-007	Every language rendering references the canonical source version. Material canonical changes mark dependent renderings stale; stale renderings cannot be published.
LNG-008	QA includes semantic consistency and rendering consistency, including text expansion, special characters, supported non-Latin scripts and RTL behaviour where applicable.

27. Canonical Report Architecture

No.	Section
1	Cover
2	Report Information
3	Contents
4	Executive Summary
5	Assessment Context
6	Assessment Scope
7	Current Position
8	Strengths to Preserve
9	Key Findings
10	Detailed Assessment
11	Connected Insights
12	Risks, Constraints & Opportunities
13	Priority Framework
14	Recommended Actions
15	Decision & Investment Considerations
16	Roadmap
17	What to Preserve
18	Further Investigation
19	Recommended Next Steps
20	Assessment Methodology & Limitations
21	Closing Page

RPT-001 The universal report narrative is Context → Understanding → Connection → Decision → Action → Transparency.

RPT-002 The report renderer uses structured approved assessment objects and MUST NOT independently re-run the assessment.

RPT-003 Portal report and downloadable PDF use the same approved canonical content source.

RPT-004 Final Executive Summary is generated/validated only after Findings, Connected Insights, Confidence, Priority, Recommendations, Decision Considerations, Roadmap, Strengths and Limitations are sufficiently established.

RPT-005 Material downstream changes invalidate affected Executive Summary content and require refresh/revalidation.

28. Report Information, Cover & Contents

RPT-006 Cover contains: IO SKY; Scan name; Organisation Name; Assessment Report; Delivered [actual publication date]; Confidential. It remains minimal and does not become a dashboard.

RPT-007 Report Language appears in Report Information, not on the cover.

RPT-008 Report Information includes appropriate traceability metadata such as Assessment ID, product, organisation, publication/delivery date, report version, methodology version and publication language.

VIEW-001	Professional Contents appears after Report Information, lists chapters with actual page numbers, and in the digital PDF provides clickable internal chapter links.
VIEW-002	PDF provides proper document outline/bookmarks for major chapters.

29. Report Design System & Visual Integrity

DSN-001	Canonical print/PDF format is A4 portrait.
DSN-002	Report page/background surface is off-white. Primary text, headings and hero/section titles are black. Default grey typography is not used as the hierarchy mechanism.
DSN-003	IO SKY Orange #F58A1F and Deep Teal #0D2D2E are controlled brand accents. The report must not become a dark website on paper.
DSN-004	Visual character: premium, editorial, spacious, architectural, controlled and confident; not generic AI PDF, dashboard export or decorative infographic pack.
DSN-005	Content MUST NOT be compressed merely to reduce page count. If a component does not fit comfortably, create a continuation/additional page or approved alternate layout.
DSN-006	No clipped text, unreadable shrinkage, cut tables, broken components or crammed findings.
DSN-007	AI determines content selection within controlled components; AI does not invent uncontrolled visual design.
DSN-008	No chart, percentage, benchmark, score, trend or quantitative comparison unless underlying assessment data legitimately supports it. Qualitative information receives qualitative representation.

30. Mandatory Visual Approval Gate

DSN-009	Before final Report Design System freeze, development/design must produce representative synthetic compositions for Cover, Executive Summary spread, Finding page/spread, Connected Insights spread and Roadmap spread.
DSN-010	These compositions must demonstrate actual proposed typography, hierarchy, spacing, off-white page surface, black text, Orange/Teal usage, icons, diagrams, headers/footers, pagination and information density.
DSN-011	IO SKY must approve the visual direction before exact typography, spacing, page grid, component dimensions and reusable tokens are frozen and rolled out.
DSN-012	Developers MUST NOT bypass this gate by selecting a final typography/layout system solely from written interpretation.

31. Portal Book Viewer & PDF

VIEW-003	Client Portal report opens as a professionally produced digital book/publication, beginning with the cover.
VIEW-004	Viewer includes Previous, Next, page indicator, Contents navigation and chapter jump; touch devices support appropriate swipe behaviour.
VIEW-005	Viewer is not a generic vertical browser PDF embed.
VIEW-006	Large displays may show one or two pages depending on usable width. Standard laptops use one page where a spread would make content too small. Tablets use one/two based on width/orientation. Mobile uses one page.

VIEW-007	Never squeeze two tiny A4 pages into a viewport merely because the device is classified as desktop.
VIEW-008	PDF supports A4, clickable Contents, internal destinations, bookmarks, page numbering, correct breaks, print-safe margins, multilingual characters, Assessment ID, Report Version and publication metadata.
VIEW-009	Published downloadable PDF is immutable/versioned and available only to the corresponding authenticated/authorised customer and authorised internal roles.
VIEW-010	Standard report motion is limited to subtle interface feedback/page navigation. No animated roadmap build-up, causal diagrams or animated storytelling. Respect reduced-motion preference.

32. Methodology/IP Boundary

Approved high-level customer-facing methodology wording:

About this assessment
This assessment combines structured organisational context, adaptive questioning and cross-answer analysis to identify relevant strengths, constraints, risks, opportunities and decision priorities. Findings are assessed in context rather than from individual answers in isolation and are subject to expert review before publication.

The assessment reflects the information available during the assessment and should be read together with any limitations identified in this report.

IP-001	Customer-facing methodology MUST NOT disclose the proprietary Question Library, branching rules, trigger thresholds, Depth logic, evaluation formulas, internal confidence calculations, prompt architecture, reasoning-framework implementation, internal reviewer rules or proprietary decision logic.
IP-002	Public preview and customer report explain enough for responsible interpretation but do not expose proprietary implementation mechanics.

33. Public Report Preview

PREV-001	Public CTA is Preview the Report, not Preview the Assessment.
PREV-002	Public preview uses synthetic/demo company data and the same real IO SKY Report Design System.
PREV-003	Public preview is limited to Cover plus one Executive Summary spread/page experience and has no full-report download.
PREV-004	Do not publicly expose full Contents, Roadmap, detailed findings, recommendation/root-cause architecture, proprietary methodology, question library, branching/depth logic, reviewer package or internal evaluation logic.

34. Executive Briefing

EB-001	Executive Briefing is a separate premium audiovisual deliverable. It is not an animated PDF, report read-aloud, generic TTS presentation or generic AI video.
EB-002	Its substantive source is exclusively the approved canonical report. It MUST NOT introduce new findings, priorities, recommendations, certainty or decision implications.

EB-003	Pipeline: Approved Canonical Report → Executive Content Selection → Executive Narrative → Storyboard → IO SKY Motion Composition → Professional Voice-over → Visualisation/Animation → Captions/Subtitles where applicable → Content/Language/Version QA → Publication Candidate.
EB-004	Narrative focuses on overall assessment, strengths, material findings, important relationships/dependencies, relevant risks/opportunities, priorities, recommended direction, decision implications and roadmap; it does not narrate every report page.
EB-005	Motion must explain sequence, relationship, hierarchy or consequence. Decorative animation, gaming aesthetics, excessive glow and generic stock 'AI' motion are prohibited.
EB-006	Voice-over must be professional, natural, calm, authoritative, correctly paced and correctly pronounce company names, technical terms and abbreviations. It must not sound robotic, overdramatic or promotional.
EB-007	Voice-over language follows Customer Publication Language.
EB-008	Briefing stores Assessment ID, canonical source version, Briefing Version, language, generation timestamp and QA status.
EB-009	A material approved-report change invalidates affected Briefing output until regenerated/revalidated against the current approved version.
EB-010	Target duration/range, scene-count guidance, subtitle/caption default, playback/streaming behaviour, download entitlement, voice configuration and final Briefing QA ownership are centrally configurable production parameters requiring IO SKY approval before production freeze; they are not to be hard-coded into prompts.

35. Authoritative State Machine

Technical names may differ, but the business states and controls represented by them may not be lost.

- PRE_PURCHASE_IN_PROGRESS
- PAYMENT_PENDING
- PAYMENT_CONFIRMED
- PAYMENT_FAILED
- ACCOUNT_ACTIVATION_PENDING
- EXISTING_ACCOUNT_BINDING
- ACCOUNT_BINDING_FAILED
- ENTITLEMENT_READY
- ASSESSMENT_IN_PROGRESS
- ASSESSMENT_SUBMITTED
- ANALYSIS_QUEUED
- ANALYSIS_IN_PROGRESS
- ANALYSIS_FAILED
- REPORT_GENERATION
- REPORT_GENERATION_FAILED
- REVIEW_PREPARATION
- REVIEW_ASSIGNMENT_REQUIRED

- REVIEW_ASSIGNED
- IN_EXPERT_REVIEW
- TARGETED_REGENERATION
- RETURNED_FOR_REANALYSIS
- REANALYSIS_IN_PROGRESS
- REVIEW_COMPLETED
- APPROVED_AWAITING_PUBLICATION
- RETURNED_BY_SUPER_ADMIN
- LANGUAGE_RENDERING
- EXECUTIVE_BRIEFING_PREPARATION
- PUBLICATION_PREPARATION
- PUBLICATION_BLOCKED
- READY_TO_PUBLISH
- PUBLICATION_IN_PROGRESS
- PUBLICATION_FAILED
- PUBLISHED

STATE-001 Every material state transition defines current state, event, permitted actor/system, required permission/scope/assignment, prerequisites, backend validations, data mutation, version impact, audit event, notification, customer-visible status, failure state and retry behaviour.

36. State Transition Matrix

From	To	Actor	Precondition	Core side effect	Notification/Audit
Pre-purchase	Payment pending	Customer	Required purchase/intake valid	Create payment attempt/order state	Payment event/audit
Payment pending	Payment confirmed	Trusted payment backend	Authoritative provider confirmation	Persist payment confirmation	Payment confirmed
Payment confirmed	Account activation pending	System - new customer	New customer path	Create secure activation eligibility	Account activation required
Payment confirmed	Existing account binding	System - authenticated client	Valid authenticated Client	Prepare binding	Binding event
Activation/binding	Entitlement ready	System	Identity + purchase + org validated	Bind account/org/purchase/entitlement	Entitlement ready
Entitlement ready	Assessment in progress	System/customer	Correct entitlement	Create/activate Assessment ID	Assessment available
Assessment in progress	Assessment submitted	Customer	Required state valid	Freeze Submitted Snapshot	Submission event
Submitted	Analysis in progress	System	Immutable snapshot exists	Start analysis version	Analysis started
Analysis complete	Review preparation	System	Canonical draft generated	Create NL/EN renderings + review intelligence	Review preparation
Review preparation	Assignment required	System	Review artifacts ready	Create assignment task	Super Admin notification
Assignment required	Review assigned	Super Admin	Eligible reviewer + scope	Bind reviewer assignment	Reviewer notification + audit
Review assigned	In expert review	Assigned reviewer	Permission/scope/assignment valid	Open review session	Review started
In expert review	Targeted regeneration	Reviewer	Target identified	Create controlled regen job	Audit change
In expert review	Returned for re-analysis	Reviewer	Mandatory reason	Create re-analysis request	Audit + task
In expert review	Review completed	Reviewer	Review gate passes	Lock review revision	Handoff notification
Review completed	Approved awaiting publication	Reviewer/System	Handoff complete	Create approved candidate	Super Admin task
Approved awaiting publication	Returned by Super Admin	Super Admin	Mandatory note	Invalidate approval for changed content	Reviewer task + audit
Approved awaiting publication	Publication preparation	System/Super Admin	Language/deliverable prerequisites	Prepare final deliverables	Publication prep event
Ready to publish	Publication in progress	Super Admin	All gates pass	Start atomic publish	Audit publish attempt
Publication in progress	Published	System	Atomic publish succeeds	Bind immutable published version + entitlement	Customer ready notification + audit

37. Concurrency, Idempotency & Stale State

OPS-001 The system MUST prevent uncontrolled conflicting edits/transitions in double Submit, double Publish, two reviewer tabs, stale reviewer revisions, Super Admin review during new revision, duplicate webhooks, translation finishing after canonical change and Briefing finishing against obsolete source.

OPS-002 A stale actor/request MUST NOT silently overwrite a newer authoritative version.

OPS-003 Idempotency must be applied to business operations where retries/duplicates could create duplicate entitlements, assessments, submissions, review jobs, publication actions or customer-visible deliverables.

38. Failure & Recovery Matrix

Failure	Persisted rule	State	Recovery	Visibility
Payment confirmation delayed	Keep pending; no entitlement	Payment pending	Reconcile/retry provider event	Internal visibility; customer sees payment processing where appropriate
Payment confirmed, browser closed	Preserve purchase	Account activation pending	Secure resume/activation	No loss of purchase
Account activation fails	Preserve paid purchase	Account activation pending/failed substate	Secure retry/recovery	Internal attention if unresolved
Duplicate webhook	Ignore duplicate business effect	No duplicate state	Idempotent acknowledge/reconcile	Audit correlation
Autosave fails	Keep last confirmed state	Assessment in progress	Retry; show Save failed	Never show Saved falsely
Analysis provider fails	Preserve Submitted Snapshot	Analysis failed	Controlled retry	No partial report visible
Report render fails	Preserve canonical content	Report generation failed	Retry renderer	No re-analysis unless required
Translation fails	Block affected language	Publication blocked	Retry rendering/QA	No stale/partial language publish
Review Brief fails	Block required review prep	Review preparation	Regenerate/recover	Do not silently omit required artifact
Executive Briefing fails	Report remains governed separately; briefing not ready	Briefing preparation failed	Retry/regenerate	Do not mark deliverable ready
Publication fails	No partial customer release	Publication failed	Controlled retry after validation	Internal alert/task
Notification fails after valid publish	Published remains valid	Published	Retry notification separately	Do not roll back valid publication solely for notification

FAIL-001 For every critical asynchronous operation, implementation documentation must define detection, persisted state, retry eligibility, safe automatic retry where applicable, manual recovery action, responsible actor, customer-visible behaviour, internal notification/task, audit event and correlation/idempotency behaviour.

39. Data Ownership, Mutability & Logical Data Model

DATA-005 Pre-submission customer answers may be editable according to assessment UX.

DATA-006 Submitted Snapshot is immutable.

DATA-007 Derived analysis may evolve only through controlled revisions.

DATA-008 Approved Canonical Report is version-controlled; material modification invalidates approval.

DATA-009 Published Version is immutable. A correction requires a new controlled version, never silent alteration of Published v1.

DATA-010 The implementation must be capable of representing the following logical distinctions. Database tables/services may differ, but these concepts may not be collapsed where doing so loses required behaviour.

- Client
- Organisation
- Organisation Membership
- Purchase/Order
- Billing Record
- Entitlement
- Assessment

- Assessment Snapshot
- Methodology Version
- Answer
- Context Object
- Signal
- Evidence State
- Investigation Path
- Contradiction
- Unknown
- Strength
- Candidate Finding
- Finding
- Finding Relationship
- Confidence
- Priority
- Recommendation
- Decision Consideration
- Roadmap Item
- Canonical Report Version
- Review Rendering
- Review Assignment
- Expert Review Brief
- Review Change
- Reviewer Note
- Review Summary
- Language Rendering
- Publication Version
- Executive Briefing Version
- Audit Event
- Notification Event

40. Business Data Dictionary

Field/Object	Meaning	Source	Mutability	Used by
Assessment ID	Permanent platform-wide Scan reference	System	No	All lifecycle resources
Methodology Version	Rules/content version governing assessment	System/config	No after assessment creation except controlled migration	Assessment
Assessment Snapshot	Assessment-specific organisation/context data	Customer/system	Editable pre-submit; immutable submitted version	Assessment
Submitted Snapshot	Frozen customer source state	System on submit	No	Analysis/review traceability
Primary Audience	Dominant report communication profile	Customer	Pre-submit; controlled correction before publish if authorised	Rendering
Secondary Audience	Supplementary profile	Customer	Same boundary	Rendering
Customer Publication Language	Final client language	Customer	Pre-submit; controlled correction before final rendering	Rendering
Review Language	Reviewer working language NL/EN	Reviewer	Review workflow	Internal only
Finding	Structured assessment conclusion	AI/reviewer	Revision controlled	Canonical report
Confidence	Qualitative certainty per finding	AI/reviewer	Revision controlled	Finding/report
Priority	Qualitative action priority	AI/reviewer	Revision controlled	Finding/roadmap
Canonical Report Version	Single approved content truth	System/reviewer	Versioned; no destructive overwrite	All output
Publication Version	Immutable delivered report version	System/Super Admin publish	Immutable	Client Portal/PDF
Delivered Date	Successful publication date	System	No manual edit	Cover/report info
Executive Briefing Version	Briefing tied to approved report version	System	Versioned	Client deliverable

41. Structured AI Output & Source of Truth

AI-009 Critical AI stages must produce application-validatable structured data. The platform MUST NOT rely solely on one long free-text model output.

AI-010 A Finding must be capable of separately representing Finding ID, type, title, basis, what was found, why it matters, contributing factors, root-cause position, business impact, Confidence/rationale, Priority/rationale, dependencies, related Findings, Recommendation, Decision Implications, Roadmap relationship and limitations.

DATA-011 Authoritative content chain: Purchase/Context → Assessment Snapshot → Structured Assessment State → Findings Registry → Relationships → Priority/Recommendations/Roadmap → Approved Canonical Report → Language/Audience Rendering → Portal/PDF → Executive Briefing.

DATA-012 Portal, PDF and Executive Briefing MUST NOT independently ask an AI provider to regenerate the underlying assessment.

42. Permissions & Actions Matrix

Action	Client	Developer Reviewer	Admin	Super Admin	System
View own in-progress assessment	Yes	No unless assigned/scope	No unless permission/scope	Yes where authorised	System N/A
Edit own pre-submit answers	Yes	No	No	No except controlled support flow	Validate/per
View submitted source answers	Own assessment where product allows portal visibility; not internal notes	Assigned + review permission	With review permission/scope	Yes	Process
View Review Brief/internal notes	No	Assigned + permission	With scope/permission	Yes	Generate
Edit Finding	No	Assigned reviewer	Assigned/authorised reviewer	Yes when reviewer	Version/cont
Request targeted regeneration	No	Assigned reviewer	Assigned/authorised reviewer	Yes	Execute
Return for re-analysis	No	Assigned reviewer	Assigned/authorised reviewer	Yes	Execute
Complete Expert Review	No	Assigned reviewer	Assigned/authorised reviewer	Yes when reviewer	Validate gat
Publish assessment	No	No	Only if explicit AI Scan Publish permission and policy allows; Super Admin retains ultimate control	Yes	Atomic publi
View another client's assessment	No	Only assigned/scope	Only permitted scope	Yes within authorised scope	Enforce
Download unpublished report	No	Assigned review scope if required	Permitted internal scope	Yes	Generate au access
Download published report	Own entitlement	Assigned/internal scope	Permitted internal scope	Yes	Serve autho

SEC-001 Access is enforced server-side by Role + Permission + Scope + Assignment/Ownership + Environment + Resource. Frontend visibility and notification never grant access.

SEC-002 Reviewer assignment never grants ownership or customer entitlement.

43. Tenant Isolation & Resource Security

SEC-003 Client A MUST NEVER access Client B resources through URL manipulation, Assessment ID, API request, Organisation ID, report ID, download endpoint, search, cached route, signed asset or manipulated request.

SEC-004 Reviewer access changes/ends when assignment, scope or permission no longer permits access.

SEC-005 Wrong report in wrong portal is a critical security/privacy failure and release blocker.

SEC-006 Publication binding, account binding, entitlement and state transitions must be validated backend-side; frontend state is not trusted as authority.

SEC-007 Customer information from one organisation MUST NOT become assessment context for another organisation.

SEC-008 Internal Review Notes, Review Brief and Internal Assessment State are not automatically customer-visible.

44. Audit Event Catalogue

Event	Domain	Actor	Minimum metadata
PURCHASE_CREATED	Purchase	Customer/System	Assessment/order reference, product, org, timestamp
PAYMENT_CONFIRMED	Billing	System/provider	Payment ref, amount/tax/currency/status, timestamp
ACCOUNT_BOUND	Identity	System	Account, organisation, purchase, entitlement references
ASSESSMENT_CREATED	Assessment	System	Assessment ID, product, methodology version
ASSESSMENT_SUBMITTED	Assessment	Customer	Assessment ID, submitted snapshot version
ANALYSIS_STARTED/COMPLETED/FAILED	Analysis	System	Assessment ID, job/version, status, correlation
REVIEWER_ASSIGNED/REASSIGNED	Review	Super Admin	Reviewer, scope, previous/new assignment, reason where applicable
MATERIAL_REVIEW_CHANGE	Review	Reviewer	Before/after reference, change type, rationale
TARGETED_REGENERATION	Review	Reviewer/System	Target, source version, result version
RETURNED_FOR_REANALYSIS	Review	Reviewer	Mandatory reason
REVIEW_COMPLETED	Review	Reviewer	Review revision, gate result
SUPER_ADMIN_RETURN	Publication	Super Admin	Mandatory note, prior approved version
PUBLICATION_ATTEMPT/SUCCESS/FAILURE	Publication	Super Admin/System	Approved version, destination binding, result
VERSION_CREATED	Versioning	System	Object/version/source version
ENTITLEMENT_CHANGED	Entitlement	Authorised actor/System	Old/new entitlement, reason

AUDIT-001 Audit history must be protected from ordinary modification/deletion.

AUDIT-002 Where applicable, audit metadata includes Assessment ID, actor/system, event, old/new state, affected resource/version, timestamp, reason and correlation/reference.

45. Notification Event Matrix

Trigger	Recipient	Family	Rule	Reference
Payment/account recovery attention	Customer + internal where required	Account/Billing	Conditional	Assessment/order ref
Assessment submitted	Customer confirmation + internal workflow	AI Scan	Required	Assessment ID
Reviewer assignment required	Super Admin	Review	Required	Assessment ID
Reviewer assigned/reassigned	Reviewer; Super Admin visibility	Review	Required	Assessment ID
Review overdue	Assigned reviewer + Super Admin	Review	Conditional/SLA	Assessment ID
Re-analysis required	Reviewer/internal workflow	Review	Required when triggered	Assessment ID
Returned by Super Admin	Reviewer	Review	Required	Assessment ID
Review completed / publication review required	Super Admin	Publication	Required	Assessment ID/version
Publication blocked/failed	Authorised internal roles	Publication	Required	Assessment ID/error ref
Publication successful / report ready	Customer + internal audit	AI Scan	Required	Assessment ID/published version

NOT-001 Notifications inform users; they do not change permissions or lifecycle state. Exact channel/template behaviour follows the IO SKY Notification Center specification.

46. Operational Visibility & Recovery

OPS-004 Authorised internal users must be able to determine without manual production-database inspection: current stage, processing failure, blocking reason, assigned reviewer, review status, overdue state, language rendering status, Executive Briefing status, publication state and permitted retry/recovery action.

OPS-005 Operational retry controls must respect idempotency, permissions and version validity. A retry may not bypass publication/review/security gates.

47. Configuration Versus Code

Item	Control	Reason
Product names/availability/prices	Config	Commercially changeable
Included deliverables/add-ons	Config	Commercially changeable
Supported languages/audience profiles	Config	Product presentation
Shared Intake wording	Config/versioned where assessment meaning changes	Customer context
Question/domain libraries	Versioned methodology config	Product-specific WHAT
Objectives/triggers/follow-up/depth metadata	Versioned methodology config	Adaptive behaviour inputs
Report labels/controlled priority labels	Config	Presentation
Notification templates	Config	Messaging
Executive Briefing availability/production parameters	Config	Deliverable
Authorisation/tenant isolation	Code/system controlled	Security
Payment verification	Code/system controlled	Financial integrity
Audit integrity	Code/system controlled	Compliance/integrity
Publication binding/immutability/state-transition security	Code/system controlled	Security/integrity

CFG-001 Business content that IO SKY reasonably needs to change without rebuilding core application logic should be data/configuration-driven where appropriate.

CFG-002 Methodology-affecting configuration changes are versioned. Historical assessments remain linked to the version that governed them.

CFG-003 Security, authorisation, tenant isolation, audit integrity, payment verification and core state-transition integrity are not ordinary editable business configuration.

48. Accessibility & Usability

A11Y-001 Assessment and report interfaces provide appropriate keyboard usability, focus states, readable contrast, responsive layouts, understandable validation/error feedback and reduced-motion support where motion exists.

A11Y-002 Adaptive behaviour must not make the customer uncertain about whether work is saved, whether input was accepted or whether the assessment is progressing.

A11Y-003 Auto-advance must not remove reasonable accessibility/control alternatives; explicit Next/Continue remains available where appropriate.

49. Synthetic Reference Scenario

Northstar Manufacturing B.V. — synthetic example only
Assessment ID: OPS-2026-000184

Starting assumption: “We need a new CRM.” The assessment investigates Sales → Operations handover and discovers inconsistent information capture, manual reconciliation and individual knowledge dependency. Existing CRM capability may support stronger configuration, so wholesale replacement is not yet established.

Candidate/established finding example: F-03 — Fragmented customer information is increasing operational dependency.

Connected insight: Inconsistent information ownership → incomplete handover → manual reconciliation → individual dependency → constrained scaling. A mitigating strength is direct cross-functional communication.

Decision direction: Optimise / Configure Before Replacement. Roadmap: NOW information ownership → NEXT handover consistency → THEN targeted integration/automation → LATER reassess genuine CRM limitations.

REF-001 This synthetic scenario validates assumption challenge, persistent memory, cross-answer connection, strength preservation, proportionate recommendation and avoidance of premature software replacement. It is not product methodology content.

50. Acceptance Test Suite

Test	Scenario	Acceptance
TST-001 Form Test	If experience is effectively customer fills long form → AI generates PDF	FAIL
TST-002 Memory Test	Early material information is used much later and after resume	PASS required
TST-003 Adaptiveness Test	Materially different organisations produce materially different relevant paths	PASS required
TST-004 Fixed Sequence Test	Hidden fixed 35-question sequence with cosmetic variation	FAIL
TST-005 Purpose Test	Every generated follow-up maps to legitimate assessment purpose/state	PASS required
TST-006 Unknown Test	I don't know remains valid unknown; no forced guessing	PASS required
TST-007 Contradiction Test	Material conflict is detected, clarified/recorded and affects state	PASS required
TST-008 Assumption Challenge	Proposed solution is investigated before treated as necessary	PASS required
TST-009 Strength Test	Relevant strengths identified and incorporated	PASS required
TST-010 Do-Not-Change Test	Can conclude preserve/monitor/no material intervention/do not replace	PASS required
TST-011 No-Scoring Test	No fabricated maturity/readiness/confidence/priority numbers	PASS required
TST-012 Finding Traceability	Reviewer can trace finding to Assessment Basis/source interactions	PASS required
TST-013 New Customer	Payment → verified activation → binding → entitlement → portal	PASS required
TST-014 Public Signup	Unrestricted direct signup attempt	MUST FAIL
TST-015 Purchase Recovery	Browser closes after paid before activation; purchase recoverable	PASS required
TST-016 Existing Client	Stays logged in, prefill works, no duplicate account/org	PASS required
TST-017 Snapshot Boundary	Assessment edit does not silently change master organisation	PASS required
TST-018 Prior Answers	Old assessment answers are not silently copied	PASS required
TST-019 Payment Idempotency	Duplicate webhook does not duplicate business objects	PASS required
TST-020 Payment Authority	Success redirect without backend confirmation grants no entitlement	PASS required
TST-021 Expert Review	Assignment, permissions, NL/EN, brief, inspector, edits, regen, reanalysis, handoff	PASS required
TST-022 Translation	Facts/certainty/findings/priority/recommendation/roadmap equivalent	PASS required
TST-023 Report/PDF	Cover, metadata, Contents, links, bookmarks, pagination, IDs/date, no fake visuals	PASS required
TST-024 Book Viewer	Large desktop/laptop/tablet/mobile readability and spread logic	PASS required
TST-025 Executive Briefing	Approved source, no new conclusions, narrative/motion/voice/language/version	PASS required
TST-026 Cross-Tenant	URL/API/ID/download/search/cache/manipulated request attempts	ALL MUST FAIL
TST-027 Concurrency	Double submit/publish, stale revisions, stale translations/briefings	PASS required
TST-028 Failure Recovery	Forced failures recover without wrong publication/corruption	PASS required
TST-029 Auto-Advance	Valid simple input saves then advances; invalid input does not; Back remains usable	PASS required
TST-030 Back-State Integrity	Changing earlier material answer updates/invalidate dependent assessment state	PASS required

51. Requirement Status & Acceptance Evidence

TRACE-001 Every normative requirement receives an implementation state: Not Started; In Development; Implemented — Awaiting Validation; Validated; Blocked; Not Applicable — Approved Reason Required.

TRACE-002 Implemented does not equal Validated. Validation requires evidence appropriate to the requirement.

TRACE-003 For each normative Requirement ID, developer handoff/acceptance must identify implementation mapping, validation method/test and evidence where applicable.

TRACE-004 Engineering traceability must work Requirement → Implementation → Test → Evidence and Test → Requirement. Assessment-content traceability must work Finding → Assessment Basis; Recommendation → supporting Finding(s); Roadmap item → relevant Recommendation/Finding/dependency where applicable.

Requirement	Component	Validation	Evidence expected	Status
ACC-003	Account Service / registration policy	TST-014	Backend test showing unrestricted signup blocked	Pending
MEM-001	Assessment State / persistence	TST-002	Long-session + resume evidence	Pending
DEP-001	Adaptive/Depth Engine	TST-003/TST-005	Scenario transcripts + structured state	Pending
REC-001	Recommendation Engine	TST-008/TST-010	Proportionality scenarios	Pending
REV-003	Review Assignment	TST-021	Assignment/permission evidence	Pending
PUB-006	Publication Binding	TST-026	Server-side binding/cross-tenant tests	Pending
SEC-003	Tenant Isolation	TST-026	Negative access suite	Pending
EB-002	Executive Briefing	TST-025	Source-version/content consistency evidence	Pending
UX-007	Auto-advance	TST-029	UX interaction test	Pending
UX-010	Back-state integrity	TST-030	Dependency invalidation/re-evaluation test	Pending

The development team must expand the traceability matrix to all normative requirements; the rows above are the minimum format and representative mapping, not permission to omit other IDs.

52. Developer Handoff Requirement

HAND-001 Development team must map this Master to its proposed/current architecture, including customer journeys, account service, organisation model, payment, entitlement, Assessment ID, Assessment State, persistence, AI/provider architecture, adaptive Engine, methodology/configuration, versioning, evidence, findings, recommendations, review, report rendering, Book Viewer, PDF, language rendering, Executive Briefing, publication, security, audit, notifications, observability, failure/retry and tests.

HAND-002 Any requirement not supported by current implementation must be marked honestly as Not Started, In Development, Awaiting Validation or Blocked rather than silently omitted or treated as implemented.

HAND-003 IO SKY is specifying product behaviour and acceptance outcomes, not performing the development team's engineering design work. Developers remain responsible for sound implementation choices and must explain material architecture/provider decisions where requested.

53. Definition of Done

- Applicable normative requirements are implemented and validated.
- All three products use one Universal Engine rather than three independent engines.

- New-customer and existing-client journeys pass.
- Payment authority, recovery and idempotency pass.
- Adaptive behaviour, persistent memory and resume pass.
- Contradiction/unknown/strength/do-not-change behaviour passes.
- No-scoring rule passes.
- Finding and recommendation traceability passes.
- Expert Review and Super Admin publication pass.
- Translation integrity passes.
- Report/PDF and Book Viewer pass.
- Executive Briefing passes where applicable.
- Tenant isolation/security passes.
- Concurrency and failure/recovery pass.
- Audit and operational visibility are functional.
- Five mandatory visual reference compositions are approved before final Report Design System token freeze.
- Complete developer traceability evidence exists.
- IO SKY accepts the resulting behaviour.

DONE-001	The system is NOT Done merely because screens exist, Stripe works, AI asks questions, a PDF generates, a reviewer can click Approve or a report appears in the Client Portal.
DONE-002	A feature is accepted only when its applicable Requirement IDs are validated against the required outcome, including negative/security/failure tests where relevant.

54. Product-Specific Methodologies That Follow

METH-001	Operations Scan Assessment Methodology is a separate specification defining Operations-specific domains, objectives, controlled question library, signals, triggers, relationships and product-specific investigation logic.
METH-002	Cyber Scan Assessment Methodology is a separate specification defining Cyber-specific WHAT.
METH-003	Elite Scan Assessment Methodology is a separate specification defining cross-domain Elite-specific WHAT and MUST NOT imply the other two products are superficial.
METH-004	All three methodologies use this Universal Engine and MUST NOT create independent engines with inconsistent memory, evidence, review, reporting, publication or security behaviour.

55. Freeze Boundaries & Controlled Dependencies

Functionally specified in this Master: customer/account journeys, payment authority, Shared Intake, assessment UX behaviour, adaptive intelligence, memory, depth, evidence, confidence, findings, relationships, priorities, recommendations, roadmaps, review, translation, report architecture, publication, security, operations, audit, failure handling and acceptance.

Intentionally not frozen as arbitrary engineering detail: exact framework/library, database-table names, queue technology, ORM, service decomposition, internal function names, provider-specific implementation mechanics.

Intentionally pending visual approval: final font family, typography scale, exact spacing/grid/component dimensions. These are derived only after the five mandatory report reference compositions are approved.

Intentionally separate: Operations, Cyber and Elite product methodologies.

Executive Briefing production configuration: duration/range, scene guidance, subtitle default, playback/download entitlement and final QA ownership remain centrally configurable approval parameters, not gaps that permit changing the core deliverable definition.

56. Final Universal Quality Standard

The final IO SKY AI Scan must behave as a premium professional assessment that understands context, remembers what the customer has said, investigates rather than merely asks, challenges assumptions without forcing a conclusion, distinguishes evidence from inference, identifies relationships and root causes, recognises strengths, tells the customer what should not change, makes proportionate recommendations, supports investment decisions, produces a professionally designed assessment publication, subjects that output to meaningful expert review, professionally renders it for the customer's audience and language, and only then securely publishes it to the correct customer.

Anything materially below this standard is not equivalent implementation merely because individual screens, API endpoints, AI calls or generated files technically function.

Appendix A — Non-Interpretation Prohibitions

- Replacing adaptive assessment with a fixed questionnaire.
- Replacing structured persistent memory with short model context only.
- Adding arbitrary numerical scoring.
- Treating customer assumptions as established conclusions.
- Recommending interventions larger than evidence supports.
- Removing mandatory human Expert Review.
- Automatically publishing AI drafts.
- Letting reviewer assignment transfer customer ownership.
- Allowing unrestricted public Client registration.
- Creating duplicate existing-client accounts/organisations.
- Silently overwriting Organisation Master Data from assessment edits.
- Copying historical assessment answers into a new Scan.
- Using client-side payment success as entitlement proof.
- Exposing one Client's resources to another.
- Allowing translation to change conclusions/certainty.
- Independently regenerating report/PDF/Briefing conclusions.
- Changing publication destination client-side.
- Publishing against stale versions.
- Silently altering Published Versions.
- Exposing proprietary methodology through public preview/report.
- Skipping mandatory visual approval gate.

- Implementing auto-advance in a way that bypasses validation/persistence or removes reasonable Back/control behaviour.

Appendix B — Pre-Freeze / Freeze-Candidate Status

This document is suitable as the universal developer implementation and acceptance basis. It deliberately leaves only controlled dependencies that should not be guessed in text: final report typography/spacing/component tokens after visual reference approval; separate product-specific methodologies; and final Executive Briefing production parameters. These dependencies do not reopen the universal product behaviour specified here.

Recommended document status: Freeze Candidate v1.0. After IO SKY approves the five report reference compositions and the separate methodology documents are aligned to this Master, the relevant sections can be promoted to final production freeze without rewriting the universal architecture.